C10.3 External snoop filter requirements

The snoop filter must ensure that it does not cause a capacity overflow by considering cache lines accessed using ReadNoSnoop and WriteNoSnoop transactions to be allocated. Such cache lines are non-shareable locations, and master components are not required to issue Evict transactions to these locations.

Snoop filters must consider speculative reads. For example, a snoop filter cannot rely on an allocating transaction to determine whether it must add a cache line to its list of allocated cache lines, and must check the current list of allocated lines so that it does not hold two copies of the same line, potentially overflowing its resources.

The snoop filter must be able to track the allocation of all cache lines that could be allocated in the associated cache. Typically, the storage within a snoop filter will match the structure of the cache for which it is filtering snoops, in terms of the associatively, the size of the cache tags, and the total number of cache lines being tracked.

Transactions in progress, and other caching structures within a master, can cause a situation where the snoop filter is required to track additional cache lines. A snoop filter can include additional storage to enable it to track these additional cache lines.

To avoid snoop filter overflow, where the tracking requirements exceed the total capabilities of the snoop filter, the snoop filter is permitted to issue snoop transactions. Transactions, such as CleanUnique, are used to remove cache lines from the associated cache. Use of this technique ensures that the snoop filter can continue to correctly track all allocated cache lines.

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Chapter C11
ACE-Lite

This chapter describes the ACE-Lite interface. It contains the following sections:

- *About ACE-Lite* on page C11-276
- *ACE-Lite signal requirements* on page C11-277.

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